

Chapter 3

Kernel Quantile Estimation



Abstract From kernel distribution function, quantile estimators can be defined naturally. Using the kernel estimator of the p -th quantile of a distribution brings about an improvement in comparison to the sample quantile estimator. The improvement is revealed when applying the Edgeworth expansion of the kernel estimator. Using one more term beyond the normal approximation significantly improves the accuracy for small to moderate samples. The investigation is nonstandard since the influence function of the resulting L -statistic explicitly depends on the sample size. We obtain the expansion, justify its validity and demonstrate the numerical gains in using it.

Quantile estimation is an important problem in statistical inference. This is mostly due to the important practical applications of the method in the financial industry and risk assessment. The risk measure VaR (value at risk) is a quantile of the population distribution. It is known that the Lorenz curve [1], as an important indicator in econometrics for income distributions, also represents the transformation of quantiles. Further, estimating the quantile itself as accurately as possible is an essential tool for inference; it is important in its own right and will be dealt with in this paper.

3.1 Quantile Estimators

For a continuous random variable X with a cumulative distribution function $F(\cdot)$, density function $f(\cdot)$ and $E|X| < \infty$, the p -th quantile is defined as $Q(p) = \inf\{x : F(x) \geq p\}$. Let $X_1, X_2, \dots, X_n$, be independently and identically distributed random variables with the distribution function $F(\cdot)$, and the simplest estimator of $Q(p)$ is the sample quantile estimator of the empirical distribution function

$$\widehat{Q}(p) = \inf\{x : F_n(x) \geq p\}$$

where $F_n(x) = \frac{1}{n} \sum_{i=1}^n I(X_i \leq x)$. It is a very popular estimator that is also typically implemented in statistical packages. Under mild conditions, it is asymptotically normal, and its asymptotic variance is $\sigma^2 = \frac{p(1-p)}{f^2(Q(p))}$ which happens to be large, par-

ticularly, in the tails of the distribution where the density $f(x)$ has small values. In risk management, the tails are important regions of interest.

An alternative estimator is the kernel quantile estimator

$$\widehat{Q}_{p,h} = \frac{1}{h} \int_0^1 F_n^{-1}(x) K\left(\frac{x-p}{h}\right) dx, \quad (3.1)$$

where $F_n^{-1}(x)$ denotes the inverse of the empirical distribution function, and $K(\cdot)$ is a suitably chosen kernel. Of course, the bandwidth should be chosen appropriately. Besides the obvious requirement $h \rightarrow 0 (n \rightarrow \infty)$, additional requirements—in combination with requirements on the kernel—must be placed to ensure consistency, asymptotic normality, asymptotic bias elimination, and higher order accuracy of the estimator $\widehat{Q}_{p,h}$.

3.2 Properties of Quantile Estimators

Let us define order statistics $X_{1:n} \leq X_{2:n} \leq \dots \leq X_{n:n}$ based on the sample $X_1, X_2, \dots, X_n$. Then the estimator $\widehat{Q}(p)$ is asymptotically equivalent to $X_{[np]+1:n}$ where $[a]$ denotes a maximum integer less than a . Further let us define order statistics $U_{1:n} \leq U_{2:n} \leq \dots \leq U_{n:n}$ based on a random sample $U_1, U_2, \dots, U_n$ from the uniform distribution $U(0, 1)$. Then $X_{[np]+1:n}$ and $F^{-1}(U_{[np]+1:n})$ are asymptotically equivalent and we can discuss asymptotic expectation and variance of $\widehat{Q}(p)$. Since the distribution of $U_{[np]+1:n}$ is a beta distribution $Be([np] + 1, n - [np])$, we can obtain the asymptotic expectation and variance as follows. Expanding around p , we have

$$\begin{aligned} E[F^{-1}(U_{[np]+1:n})] &= E\left[F^{-1}(p) + (U_{[np]+1:n} - p)\{f(F^{-1}(p))\}^{-1} \right. \\ &\quad \left. - \frac{1}{2}(U_{[np]+1:n} - p)^2 \frac{f'(F^{-1}(p))}{f^3(F^{-1}(p))}\right] + O(n^{-2}) \\ &= F^{-1}(p) + \frac{1}{f(F^{-1}(p))} E[U_{[np]+1:n} - p] \\ &\quad - \frac{1}{2} \frac{f'(F^{-1}(p))}{f^3(F^{-1}(p))} E[(U_{[np]+1:n} - p)^2] + O(n^{-2}). \end{aligned}$$

Thus, using the moments of the beta distribution, we can show that

$$E[F^{-1}(U_{[np]+1:n})] = F^{-1}(p) + \frac{A}{f(F^{-1}(p))} - \frac{1}{2} \frac{f'(F^{-1}(p))}{f^3(F^{-1}(p))} B + O(n^{-2})$$

where

$$A = \frac{[np] + 1}{n + 1} - p,$$

$$B = \frac{([np] + 2)([np] + 1)}{(n + 2)(n + 1)} - 2p \left(\frac{[np] + 1}{n + 1} \right) + p^2.$$

Since $|[np] - np| \leq 1$ and $F^{-1}(p) = Q(p)$, we have that $A = O(n^{-1})$ and $B = O(n^{-1})$. Then the bias of $\widehat{Q}(p)$ is $O(n^{-1})$. Similarly, we can obtain the variance of the estimator $\widehat{Q}(p)$

$$V(\widehat{Q}(p)) = \frac{p(1-p)}{nf^2(F^{-1}(p))} + O(n^{-2}), \quad (3.2)$$

and then an asymptotic mean squared error with residual term $O(n^{-2})$ is given by

$$AMSE(\widehat{Q}(p)) = AMSE(F^{-1}(U_{[np]+1;n})) = \frac{p(1-p)}{f^2(F^{-1}(p))n} = \frac{Q'(p)^2 p(1-p)}{n}. \quad (3.3)$$

The estimator in (3.1) has been studied by many researchers in the past (see [2–6] and the references therein). Clearly, $\widehat{Q}_{p,h}$ is an L -estimator since it can be written as a weighted sum of the order statistics $X_{i:n}$, $i = 1, 2, \dots, n$:

$$\widehat{Q}_{p,h} = \sum_{i=1}^n v_{i,n} X_{i:n}, \quad v_{i,n} = \frac{1}{h} \int_{\frac{i-1}{n}}^{\frac{i}{n}} K\left(\frac{x-p}{h}\right) dx. \quad (3.4)$$

The difficulty in using standard asymptotic theory about L -statistics when analyzing the behavior of $\widehat{Q}_{p,h}$ is related to the fact that the weights $v_{i,n}$ in (3.4) depend explicitly on n in a peculiar way and tend to vanish asymptotically. Standard statements about asymptotic expansions of L -statistics (e.g., [1, 7]) are related to the case where the L -statistic could be written as $\int_0^1 F_n^{-1}(u) J(u) du$ with the *score function* $J(u)$ (which itself is an asymptotic limit) that does *not* involve n . The reason is that it is impossible to write its “score function” in such a way since it becomes a delta function in the limit. The treatment is technically involved and this has slowed down the research on the properties of $\widehat{Q}_{p,h}$ as an estimator of $Q(p)$. It is particularly interesting to derive higher order expansions for the asymptotic distribution of $\widehat{Q}_{p,h}$. Indeed, research by [2, 6] shows that the kernel quantile estimator is asymptotically equivalent to the empirical quantile estimator up to first order, hence any advantage would be revealed by involving at least one more term in the Edgeworth expansion and for moderate sample sizes. In this paper, we will derive such an expansion, will justify its validity, and will illustrate its advantages numerically. Since the empirical distribution $F_n(x)$ is a discrete random variable, we cannot prove the validity of its formal Edgeworth expansion.

A standard approach in the study of asymptotic properties of L -statistics is to first decompose them into an U -statistic plus a small-order remainder term and to apply asymptotic theory for the main term, that is, the U -statistic. When the $J(u)$ function

depends on n , however, such a decomposition leads to an object that is similar to an asymptotic U -statistic [8] yet its defining function $h(\cdot, \cdot)$ explicitly depends on n and standard results about U -statistics can not be used directly. Progress in this direction has been achieved by [5, 6]. By applying Esseen's smoothing lemma ([9] Chap. XVI) and exploiting a nice decomposition by Friedrich for the resulting statistic, he has shown that a Berry-Esseen-type result of the form

$$P(\sqrt{n}|\hat{Q}_{p,h} - Q(p)| \leq x\sigma_n) = 2\Phi(x) - 1 + O(n^{-r}) \quad (3.5)$$

holds whereby σ_n^2 can be given explicitly in terms of the kernel K and the derivative $Q'(p)$ of $Q(p)$. The rate r depends on the order m of the kernel. When $m = 2$ we obtain $O(n^{-1/3})$, which is improved to $O(n^{-5/13})$ for $m = 3$. Naturally, for no kernel order m is it possible to get the reminder order in (3.5) down to $O(n^{-1/2})$, not to mention $o(n^{-1/2})$ without further including the next term from an Edgeworth expansion. In [10] (Lemma 1), the authors have attempted to derive such an expansion in order to improve (3.5). However, the result as stated there is incorrect. Both the definitions of the expansion function ($G_n(x)$) and the order of the expansion ($o(n^{-1/3})$) do not make sense and the argument used in the derivation is imprecise. Indeed, the authors rely on a technique about U -statistics used by [11] but in the latter paper the function $h(\cdot, \cdot)$ in the U -statistic's definition does *not* depend on n .

The purpose of this paper is to suggest a correct Edgeworth expansion for the kernel-based quantile estimator $\hat{Q}_{p,h}$ up to order $o(n^{-1/2})$. The derivation is non-trivial. The specific requirement on the bandwidth h for the sake of eliminating the asymptotic bias triggers the need to include further contributions from the terms of order $O(n^{-1})$ of the appropriately resulting U -statistic. These contributions have to be taken into account in order to achieve the desired Edgeworth expansion with remainder of order $o(n^{-1/2})$. Although the general result is involved, the expansion can be simplified significantly in the most typical and practically relevant case of a symmetric compactly supported kernel $K(\cdot)$. Finally, we present numerical evidence about the improved accuracy of the Edgeworth expansion in comparison to the normal approximation for moderate sample sizes and some common distributions $F(x)$.

The proofs of our results require novel dedicated approaches. After an extensive search of the literature, we were indeed able to find one recent result (Lemma 2.1 of [12]) concerning the Edgeworth expansion of uniformly integrable transformations of independent identically distributed random variables in the form

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n V_n(X_i) + \frac{1}{n^{3/2}} \sum_{1 \leq i < j \leq n} W_n(X_i, X_j), n \geq 2. \quad (3.6)$$

whereby the transformations $V_n(\cdot)$ and $W_n(\cdot, \cdot)$ are explicitly allowed to depend on n . For the sake of completeness, we quote the statement of lemma:

Let $W_n(x, y)$ be a symmetric function in its arguments. Assume also that:

(a) $EV_n(X_1) = 0$, $EV_n^2(X_1) = 1$, $|V_n(X_1)|^3$ are uniformly integrable and the distribution of $V_n(X_1)$ is non-lattice for all sufficiently large n .

(b) $E(W_n(X_1, X_2)|X_1) = 0$, $|W_n(X_1, X_2)|^{5/3}$ are uniformly integrable and $|W_n(X_i, X_j)| \leq n^{3/2}$ for all $n \geq 2$ $i \neq j$. Then, as $n \rightarrow \infty$,

$$\sup_x \left| P \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n V_n(X_i) + \frac{1}{n^{3/2}} \sum_{1 \leq i < j \leq n} W_n(X_i, X_j) \leq x \right) - F_n^{(1)}(x) \right| = o(n^{-1/2}), \quad (3.7)$$

where $F_n^{(1)}(x) = \Phi(x) - \frac{\phi^{(3)}(x)}{6\sqrt{n}} (EV_n^3(X_1) + 3EV_n(X_1)V_n(X_2)W_n(X_1, X_2))$. Here, $\Phi(x)$ denotes the cdf of the standard normal distribution, $\phi(x)$ is its density, and $\phi^{(3)}(x)$ is the third derivative.

However, as already pointed out, the above Lemma does not quite serve our purposes since we need one more term in the expansion of the U-statistic.

3.3 Asymptotic Properties of Quantile Estimators

For simplicity of the exposition, we will be using a compactly supported kernel $K(x)$ on $(-1, 1)$. We will say that the kernel is of order m if

$$K(x) \in L^2(-\infty, \infty), K^{(m)}(x) \in Lip(\alpha) \text{ for some } \alpha > 0,$$

$$\int_{-1}^1 K(x)dx = 1, \int_{-1}^1 x^i K(x)dx = 0, i = 1, 2, \dots, m-1, \int_{-1}^1 x^m K(x)dx \neq 0$$

For now, we do not require symmetry of the kernel.

Achieving a fine bias-variance trade-off for the kernel-based estimator is a delicate matter. It will be seen in the proof that the proper choice of bandwidth is given by

$$h = o(n^{-1/4}) \quad \text{and} \quad \lim_{n \rightarrow \infty} (n^{1/4}h)^{-k} n^{-\beta} = 0 \quad (3.8)$$

for any $\beta > 0$ and integer k . The typical bandwidth we choose to work with in the numerical implementation is $h = n^{-1/4}(\log n)^{-1}$.

Before formulating the main results of the paper, we briefly review the moment evaluations of the H -decomposition. For independent identically distributed random variables $X_1, \dots, X_n$ and a function $v(x_1, \dots, x_r)$ which is symmetric in its arguments with the property $E[v(X_1, \dots, X_r)] = 0$, we define

$$\begin{aligned} \rho_1(x_1) &= E[v(x_1, X_2, \dots, X_r)], \\ \rho_2(x_1, x_2) &= E[v(x_1, x_2, \dots, X_r)] - \rho_1(x_1) - \rho_1(x_2), \dots, \end{aligned}$$

and

$$\rho_r(x_1, x_2, \dots, x_r) = v(x_1, x_2, \dots, x_r) - \sum_{k=1}^{r-1} \sum_{C_{r,k}} \rho_k(x_{i_1}, x_{i_2}, \dots, x_{i_k})$$

where $\sum_{C_{r,k}}$ indicates that the summation is taken over all integers $i_1, \dots, i_k$ satisfying $1 \leq i_1 < \dots < i_k \leq r$. Then it holds

$$E[\rho_k(X_1, \dots, X_k) | X_1, \dots, X_{k-1}] = 0 \text{ a.s.} \quad (3.9)$$

and

$$\sum_{C_{n,r}} v(X_{i_1}, \dots, X_{i_r}) = \sum_{k=1}^r \binom{n-k}{r-k} A_k \quad (3.10)$$

where

$$A_k = \sum_{C_{n,k}} \rho_k(X_{i_1}, \dots, X_{i_k}). \quad (3.11)$$

Using Eq. (3.9) and the moment evaluations of martingales [13], we have the upper bounds of the absolute moments of A_k . For $q \geq 2$, if $E|v(X_1, \dots, X_r)|^q < \infty$, there exists a positive constant C , which may depend on v and F but not on n , such that

$$E|A_k|^q \leq Cn^{qk/2} E|\rho_k(X_{i_1}, \dots, X_{i_k})|^q. \quad (3.12)$$

Since the empirical distribution $F_n(u)$ is a sum of independent identically distributed (i.i.d.) random variables, we have moment evaluations of $|F_n(u) - F(u)|$. For $q \geq 2$, we get

$$E|F_n(u) - F(u)|^q \leq Cn^{-q/2} F(u)(1 - F(u)) \quad (3.13)$$

where C is a constant. Hereafter, C will denote a generic constant that may change its meaning at different places in the text.

Using the inequalities (3.12) and (3.13), we will obtain an asymptotic representation of the standardized quantile estimator with residual $o_L(n^{-1/2})$

$$P(|o_L(n^{-1/2})| \geq \gamma_n n^{-1/2}) = o(n^{-1/2})$$

where $\gamma_n \rightarrow 0$ as $n \rightarrow \infty$.

Let $\{Y_i\}_{i=1, \dots, n}$ be independent random variables uniformly distributed on $(0, 1)$ and define

$$\begin{aligned}
\bar{Q}(p) &= \frac{1}{h} \int_0^1 F^{-1}(x) K\left(\frac{x-p}{h}\right) dx, \\
\hat{I}_x(Y_1) &= I(Y_1 \leq p + hx) - (p + hx), \\
g_{1n}(Y_1) &= - \int_{-1}^1 Q'(p + hx) K(x) \hat{I}_x(Y_1) dx, \\
\sigma_n^2 &= \text{Var}(g_{1n}(Y_1)), \\
d_{1n} &= \sigma_n^{-1} n^{-1/2}, \quad d_{2n} = \sigma_n^{-1} n^{-3/2} h^{-1}, \quad d_{3n} = \sigma_n^{-1} n^{-5/2} h^{-2}, \\
g_{2n}(Y_1, Y_2) &= - \int_{-1}^1 Q'(p + hx) K'(x) \hat{I}_x(Y_1) \hat{I}_x(Y_2) dx, \\
g_{3n}(Y_1, Y_2, Y_3) &= - \int_{-1}^1 Q'(p + hx) K^{(2)}(x) \hat{I}_x(Y_1) \hat{I}_x(Y_2) \hat{I}_x(Y_3) dx, \\
\hat{g}_{1n}(Y_1) &= - \frac{1}{2} \int_{-1}^1 Q'(p + hx) K^{(2)}(x) E[\hat{I}_x^2(Y_2)] \hat{I}_x(Y_1) dx, \\
A_{1n} &= \sum_{i=1}^n g_{1n}(Y_i), \quad A_{2n} = \sum_{C_{n,2}} g_{2n}(Y_i, Y_j), \quad A_{3n} = \sum_{C_{n,3}} g_{3n}(Y_i, Y_j, Y_k)
\end{aligned}$$

and $\hat{A}_{1n} = \sum_{i=1}^n \hat{g}_{1n}(Y_i)$. Then we have the following lemma.

Lemma 3.1 Assume $\int [F(x)(1-F(x))]^{1/5} dx < \infty$. Let $Q^{(m)}$ be uniformly bounded in a neighborhood of p ($0 < p < 1$) and $f(Q(p)) > 0$. Let $K(x)$ be a fourth-order kernel (i.e., $m = 4$) and $K^{(4)}(x) \in \text{Lip}(\alpha)$, $\alpha > 0$. Denote $\delta = Q'(p)(\frac{1}{2} - p) + \frac{1}{2} Q^{(2)}(p)p(1-p)$. Further, choose h satisfying (3.8). Then we have

$$\begin{aligned}
&\sigma_n^{-1} \sqrt{n} (\hat{Q}_{p,h} - \bar{Q}(p)) \\
&= d_{1n} A_{1n} + d_{2n} A_{2n} + d_{3n} A_{3n} + d_{3n} (n-1) \hat{A}_{1n} + \frac{\delta}{\sigma \sqrt{n}} + o_L(n^{-1/2}).
\end{aligned}$$

The lemma shows that in order to obtain the Edgeworth expansion of $\sigma_n^{-1} \sqrt{n} (\hat{Q}_{p,h} - \bar{Q}(p))$, we can concentrate first on the Edgeworth expansion of

$$d_{1n} A_{1n} + d_{2n} A_{2n} + d_{3n} A_{3n} + d_{3n} (n-1) \hat{A}_{1n}$$

with residual term $o_L(n^{-1/2})$. We will also prove the validity of the Edgeworth expansion.

Theorem 3.1 Under the assumptions of Lemma 3.1, we have

$$P(\sqrt{n}(\hat{Q}_{p,h} - \bar{Q}(p)) \leq x\sigma_n) = G_n\left(x - \frac{\delta}{\sigma \sqrt{n}}\right) + o(n^{-1/2}). \quad (3.14)$$

Here:

$$G_n(x) = \Phi(x) - \phi(x) \left\{ \frac{x^2 - 1}{6n^{1/2}\sigma_n^3} \left(e_{1n} + \frac{3e_{2n}}{h} \right) + \frac{1}{nh^2} \left(\frac{x}{4\sigma_n^2} \{4e_{5n} + e_{6n}\} + \frac{x^3 - 3x}{6\sigma_n^4} \{3e_{2n} + e_{4n}\} + \frac{x^5 - 10x^3 + 15x}{8\sigma_n^6} e_{2n}^2 \right) \right\}.$$

where

$$\begin{aligned} e_{1n} &= E[g_{1n}^3(Y_1)], & e_{2n} &= E[g_{1n}(Y_1)g_{1n}(Y_2)g_{2n}(Y_1, Y_2)], \\ e_{3n} &= E[g_{1n}(Y_2)g_{1n}(Y_3)g_{2n}(Y_1, Y_2)g_{2n}(Y_1, Y_3)], \\ e_{4n} &= E[g_{1n}(Y_1)g_{1n}(Y_2)g_{1n}(Y_3)g_{3n}(Y_1, Y_2, Y_3)], \\ e_{5n} &= E[g_{1n}(Y_1)\hat{g}_{1n}(Y_1)] \quad \text{and} \quad e_{6n} = E[g_{2n}^2(Y_1, Y_2)]. \end{aligned}$$

Expanding $G_n(x - \frac{\delta}{\sigma\sqrt{n}})$ around x and keeping the $O(\frac{1}{\sqrt{n}})$ terms only, we can also write (3.14) as follows:

$$\begin{aligned} P(\sqrt{n}(\hat{Q}_{p,h} - \bar{Q}(p)) \leq x\sigma_n) &= \Phi(x) - \phi(x) \left\{ \frac{x^2 - 1}{6n^{1/2}\sigma_n^3} \left(e_{1n} + \frac{3e_{2n}}{h} \right) + \frac{1}{nh^2} \left(\frac{x}{4\sigma_n^2} \{4e_{5n} + e_{6n}\} + \frac{x^3 - 3x}{6\sigma_n^4} \{3e_{2n} + e_{4n}\} + \frac{x^5 - 10x^3 + 15x}{8\sigma_n^6} e_{2n}^2 \right) \right\} \\ &\quad - \frac{\delta}{\sigma\sqrt{n}}\phi(x) + o(n^{-1/2}) \end{aligned}$$

The obtained Edgeworth expansion (3.14) is not for the ultimate quantity of interest $\sqrt{n}(\hat{Q}_{p,h} - Q(p))$. Representing

$$\sqrt{n}(\hat{Q}_{p,h} - Q(p)) = \sqrt{n}(\hat{Q}_{p,h} - \bar{Q}(p)) + d_n$$

with $d_n = \sqrt{n}(\bar{Q}(p) - Q(p)) = \sqrt{n}(\frac{1}{h} \int_0^1 F^{-1}(x)K(\frac{x-p}{h})dx - Q(p))$ we need to make sure that the asymptotic order of the bias d_n is kept under control. Substituting $x - p = yh$ and applying Taylor expansion up to order m of $Q(x) = F^{-1}(x)$ around p , we get

$$d_n = O(\sqrt{nh}^m) \int_{-1}^1 K(y)y^m dy.$$

Hence

$$\begin{aligned}
P(\sqrt{n}(\widehat{Q}_{p,h} - Q(p)) \leq x\sigma_n) &= G_n\left(x - \frac{\delta}{\sigma\sqrt{n}}\right) \\
&= P(\sqrt{n}(\widehat{Q}_{p,h} - \bar{Q}(p)) \leq \left(x - \frac{d_n}{\sigma_n}\right)\sigma_n) = G_n\left(x - \frac{\delta}{\sigma\sqrt{n}}\right) \\
&= G_n\left(x - \frac{d_n}{\sigma_n} - \frac{\delta}{\sigma\sqrt{n}}\right) = G_n\left(x - \frac{\delta}{\sigma\sqrt{n}}\right) + o(n^{-1/2})
\end{aligned} \tag{3.15}$$

Expanding further G_n around the point $x - \frac{\delta}{\sigma\sqrt{n}}$, we finally get

$$P(\sqrt{n}(\widehat{Q}_{p,h} - Q(p)) \leq x\sigma_n) - G_n\left(x - \frac{\delta}{\sigma\sqrt{n}}\right) = \phi\left(x - \frac{\delta}{\sigma\sqrt{n}}\right)\left(-\frac{d_n}{\sigma_n}\right) + o(n^{-1/2}) \tag{3.16}$$

To keep the contribution of d_n under control, we require $d_n = o(n^{-1/2})$, that is, $\sqrt{nh^m} = o(n^{-1/2})$. We see now that the requirements $h = o(n^{-1/4})$, $m = 4$ that we put for another reason in Theorem 3.1, also guarantee that $d_n = o(n^{-1/2})$ holds. This is one more argument in favor of the choice $h = o(n^{-1/4})$ for the bandwidth. Moreover, the discussion in [4] (p. 411) also demonstrates from another point of view is that a reasonable choice of h should satisfy the order requirement $h = o(n^{-1/4})$. Their argument is that otherwise the dominant term in the MSE expansion of the kernel-quantile estimator may become worse than the one for the empirical quantile estimator, that is, $\frac{p(1-p)}{n}(Q'(p))^2$.

The above discussion together with Theorem 3.1, leads us to the following:

Theorem 3.2 *Under the assumptions of Theorem 3.2, we have the following Edgeworth expansion with remainder $o(n^{-1/2})$:*

$$P(\sqrt{n}(\widehat{Q}_{p,h} - Q(p)) \leq x\sigma_n) = G_n\left(x - \frac{\delta}{\sigma\sqrt{n}}\right) + o(n^{-1/2}) \tag{3.17}$$

Remark 3.1 Our expansion also leads to a Berry-Esseen bound

$$P(\sqrt{n}|\widehat{Q}_{p,h} - Q(p)| \leq x\sigma_n) = 2\Phi(x) - 1 + O(n^{-1/2})$$

This is an improvement on the result found by [5] whose bound for $m = 4$ only implies $P(\sqrt{n}|\widehat{Q}_{p,h} - Q(p)| \leq x\sigma_n) = 2\Phi(x) - 1 + O(n^{-7/17})$.

Remark 3.2 A more careful examination of the proof of Theorem 3.1 shows that for its statement to hold, one does not actually need the moment conditions $\int_{-1}^1 x^i K(x)dx = 0$, $i = 1, 2, \dots, m-1$ on the kernel $K(x)$ to hold. However, these conditions are required in order to get the order of the bias right so that the ultimate expansion given in Theorem 3.2 could be obtained.

Remark 3.3 It is known that in terms of first-order performance *with respect to MSE* the kernel quantile estimator can only match the sample quantile [4]. The improvement with respect to the sample quantile can only show up in higher order

terms of the MSE approximation (this phenomenon has been called *deficiency*). Much of the works of [2, 10] and others has been directed towards showing advantages of the kernel quantile estimator with respect to the deficiency criterion. In this paper, we approach the issue from another point of view by deriving the Edgeworth expansion for the non-lattice distribution of the kernel quantile estimator. This represents an alternative way of getting higher-order improvement in comparison to the sample quantile when estimating the quantile $Q(p)$. The Edgeworth expansion we suggest in this paper is more accurate than the normal approximation of the sample quantile and the normal approximation of the kernel quantile estimator.

Remark 3.4 Finally, we will comment on a crucial special case with important implications about the implementation of the Edgeworth expansion (3.17). We note that the general statement of Theorem 3.2 does not require *symmetry* (around zero) of the kernel $K(x)$. However, most of the kernels used in practical applications are usually symmetric [14] (p. 13). The symmetry assumption is also automatically made in Sheather and Marron [4] and in other influential papers on kernel quantile estimation. If we do assume that in addition the kernel *is* symmetric around zero (that is, $K(-x) = K(x)$) holds, then obviously $\int_{-1}^1 K'(x)dx = 0$ holds. In that case, via Taylor expansion around p , we can see that the terms e_{2n} and e_{6n} in the expansion in Theorems 3.1 and 3.2 are of smaller order in comparison to the remaining terms and we get in this case the simpler expression:

$$\begin{aligned} P(\sqrt{n}(\widehat{Q}_{p,h} - Q(p)) \leq x\sigma_n) &= \Phi(x) - \phi(x) \left\{ \frac{x^2 - 1}{6n^{1/2}\sigma_n^3} \left(e_{1n} + \frac{3e_{2n}}{h} \right) \right. \\ &\quad \left. + \frac{1}{nh^2} \left(\frac{xe_{5n}}{\sigma_n^2} + \frac{x^3 - 3x}{6\sigma_n^4} e_{4n} \right) \right\} - \frac{\delta}{\sigma\sqrt{n}} \phi(x) + o(n^{-1/2}) \end{aligned} \quad (3.18)$$

For higher order kernels like the ones we use in this paper, it is possible to make them also satisfy the additional condition $\int_{-1}^1 K''(x)dx = 0$. If such kernel is chosen then expression (3.18) simplifies even further and becomes

$$\begin{aligned} P(\sqrt{n}(\widehat{Q}_{p,h} - Q(p)) \leq x\sigma_n) &= \Phi(x) - \phi(x) \frac{x^2 - 1}{6n^{1/2}\sigma_n^3} \left(e_{1n} + \frac{3e_{2n}}{h} \right) \\ &\quad - \frac{\delta}{\sigma\sqrt{n}} \phi(x) + o(n^{-1/2}). \end{aligned} \quad (3.19)$$

Remark 3.5 We see the potentially most relevant practical applications of expansion (3.19) in evaluating power of tests of the type $H_0 : Q(p) = q_0$ against one-sided or two-sided alternatives. If the test is based on the statistic $\frac{\sqrt{n}(\widehat{Q}_{p,h} - q_0)}{\sigma_n}$ then by using the approximation (3.19) one should be able to get better power approximation for such a test in comparison to just using the normal approximation. Hereby, the approximation would be needed over the whole range of its values. Another application is in constructing more accurate confidence interval for the quantile $Q(p)$ when the sample size is small to moderate. For a given level α , instead of constructing it in a symmetric way as $\widehat{Q}_{p,h} \pm z_{\alpha/2}\sigma_n/\sqrt{n}$ one can improve the coverage accuracy

by using $(\widehat{Q}_{p,h} + c_{1-\alpha/2}\sigma_n/\sqrt{n}, \widehat{Q}_{p,h} + c_{\alpha/2}\sigma_n/\sqrt{n})$ with the quantile values $c_{1-\alpha/2}$ and $c_{\alpha/2}$ obtained from the Edgeworth approximation.

3.4 Numerical Comparisons

Our goal in this section is to demonstrate numerically the effect of the improvement in the approximation of the distribution of the kernel quantile estimator when we move from the normal to the Edgeworth approximation. This effect could be seen for moderate sample sizes such as $n = 15, 30, 40, 50$ since for very large n the two approximations would become very close to each other. We could choose different kernels satisfying the requirements of Theorem 3.2 but the effect of the kernel is not that crucial. Here, we only present results obtained with the following symmetric fourth-order kernel first suggested by H. Müller:

$$K(x) = \frac{315}{512} (11x^8 - 36x^6 + 42x^4 - 20x^2 + 3) I(|x| \leq 1).$$

The “asymptotically correct” bandwidth $h = n^{-1/4}(\log n)^{-1}$ turned out to be very well adjusted for samples of size $n \geq 50$. For smaller sample sizes (for which the $\log n$ is not small enough), it may be necessary to choose a smaller bandwidth, e.g., $0.1n^{-1/4}$, specifically when the value of p is near 0 or 1, to protect against a bias associated with edge effects.

The integral of the second derivative of this kernel is equal to zero and hence the approximation (3.19) can be applied when using it. The effect of the improvement from using Edgeworth expansion instead of the normal approximation depends on the particular distribution, on the sample size, and on the value of p . It is not to be expected to have a significant improvement when the underlying distribution is normal. Despite this, we also include the normal case in our simulations. More spectacular improvement could be expected for more skewed distributions, for smaller values of n (e.g., 15, 30), and for values of p closer to 0 or to 1. The subsections about the exponential and gamma distribution confirm this statement.

3.4.1 Estimation of Quantiles of the Standard Normal Distribution

In this case, we can derive easily:

$$Q'(p) = \frac{1}{\phi(\Phi^{-1}(p))}, \quad Q''(p) = \frac{\Phi^{-1}(p)}{\phi^2(\Phi^{-1}(p))}$$

We include the numerical values of “true” cumulative distribution function of the standardized random variable $\sqrt{n}(\widehat{Q}_{p,h} - Q(p))/\sigma_n$ and compare these values to the cumulative distribution of the standard normal and to the Edgeworth approximation as derived in Theorem 3.2. Table 3.1 gives the numerical values for two different scenarios: $n = 50$ and $n = 40$, both applied for the same value of $p = 0.1$. The “true” values of the cumulative distribution function were calculated on the basis of the empirical proportions for 2,000,000 repeated simulations. We found that at this number of simulations, the empirical ratios virtually do not change uniformly over the whole range $(-2.5, 2.5)$ of values of the argument up to the fourth decimal place. The values of e_{in} were calculated via averaging the numerical values of the resulting $g_{1n}(Y_1)$, $g_{2n}(Y_1, Y_2)$ values for 500,000 simulated independent uniform $(0,1)$ values Y_1, Y_2 , whereas σ_n was calculated as the empirical standard deviation estimator for $\sqrt{Var(g_{1n}(Y_1))}$ using the 500,000 simulated Y_1 observations. Again, the 500,000

Table 3.1 Normal sample

v	Normal	Edgeworth	True value	Edgeworth	True
-2.4000	0.0082	0.0097	0.0091	0.0111	0.0091
-2.2000	0.0139	0.0155	0.0146	0.0174	0.0146
-2.0000	0.0228	0.0243	0.0227	0.0265	0.0227
-1.8000	0.0359	0.0370	0.0349	0.0392	0.0347
-1.6000	0.0548	0.0547	0.0519	0.0568	0.0515
-1.4000	0.0808	0.0787	0.0752	0.0802	0.0746
-1.2000	0.1151	0.1103	0.1061	0.1108	0.1052
-1.0000	0.1587	0.1505	0.1457	0.1496	0.1445
-0.8000	0.2119	0.1999	0.1944	0.1973	0.1936
-0.6000	0.2743	0.2586	0.2524	0.2542	0.2520
-0.4000	0.3446	0.3257	0.3192	0.3197	0.3194
-0.2000	0.4207	0.3998	0.3934	0.3927	0.3940
0.0000	0.5000	0.4783	0.4721	0.4708	0.4737
0.2000	0.5793	0.5583	0.5530	0.5512	0.5556
0.4000	0.6554	0.6366	0.6321	0.6306	0.6359
0.6000	0.7257	0.7101	0.7069	0.7056	0.7116
0.8000	0.7881	0.7762	0.7742	0.7735	0.7795
1.0000	0.8413	0.8332	0.8320	0.8322	0.8374
1.2000	0.8849	0.8802	0.8797	0.8807	0.8852
1.4000	0.9192	0.9172	0.9172	0.9187	0.9222
1.6000	0.9452	0.9451	0.9454	0.9472	0.9495
1.8000	0.9641	0.9651	0.9656	0.9674	0.9687
2.0000	0.9772	0.9788	0.9793	0.9810	0.9815
2.2000	0.9861	0.9877	0.9880	0.9896	0.9896
2.4000	0.9918	0.9933	0.9934	0.9947	0.9945

simulations were chosen on the basis of our experimentation showing that the resulting estimated parameter values for e_{in} , and for σ_n were virtually unchanged by a further increase in the number of simulations.

Table 3.1 shows the size of the improvement and demonstrates that except for a very tiny region of values in the lower tails of the distribution, the Edgeworth expansion approximated much better the true cumulative distribution function's values. In the small region of values in the lower tail where the opposite happened, the difference in the two approximations is negligibly small (it is in the third digit after the decimal point only, whereas, in the region where the Edgeworth expansion is better, the effect of the improvement is typically observed in the second digit).

3.4.2 Estimation of Quantiles of the Standard Exponential Distribution

The following calculations are easy to derive in this case:

$$Q(p) = -\ln(1 - p), Q'(p) = \frac{1}{1 - p}, Q''(p) = \frac{1}{(1 - p)^2}.$$

Here, the improvement using the Edgeworth approximation is much more significant and can be felt for sample sizes as small as $n = 15$. Moreover, the improvement seems to be *virtually uniform* all over the whole range of values of the argument (except for a small region in the very left tail where, as is known, the Edgeworth expansion may become negative, in which case it should be set equal to zero). We include here the numerical values of “true” cumulative distribution function of the standardized random variable $\sqrt{n}(\hat{Q}_{p,h} - Q(p))/\sigma_n$ and compare these values to the cumulative distribution of the standard normal and to the Edgeworth in Table 3.2. The numerical values are given for two different scenarios: $n = 30$ and $n = 15$, both applied for the same value of $p = 0.9$. Again, the “true” values of the cumulative distribution was calculated on the basis of the empirical proportions for 2,000,000 repeated simulations. The values of e_{in} were calculated via averaging the numerical values of the resulting $g_{1n}(Y_1)$, $g_{2n}(Y_1, Y_2)$ values for 500,000 simulated independent uniform (0,1) pairs of values Y_1, Y_2 , whereas σ_n was calculated as the empirical standard deviation estimator for $\sqrt{\text{Var}(g_n(Y_1))}$ using the 500,000 simulated Y_1 observations.

Table 3.2 shows that the effect of the improvement from using the Edgeworth expansion is virtually over the whole range of values of the argument.

3.4.3 Estimation of Quantiles of the Gamma Distribution

There is a very reliable algorithm DM for calculating the inverse of the incomplete Gamma function. This alleviates a lot the testing of our approximation for any Gamma

Table 3.2 Exponential sample

v	Normal	Edgeworth	True value	Edgeworth	True
-2.4000	0.0082	-0.0052	0.0004	-0.0099	0.0000
-2.2000	0.0139	-0.0035	0.0015	-0.0095	0.0003
-2.0000	0.0228	0.0017	0.0045	-0.0055	0.0020
-1.8000	0.0359	0.0124	0.0117	0.0042	0.0078
-1.6000	0.0548	0.0307	0.0256	0.0223	0.0218
-1.4000	0.0808	0.0590	0.0494	0.0513	0.0483
-1.2000	0.1151	0.0989	0.0854	0.0930	0.0905
-1.0000	0.1587	0.1513	0.1351	0.1482	0.1487
-0.8000	0.2119	0.2155	0.1976	0.2161	0.2212
-0.6000	0.2743	0.2896	0.2712	0.2941	0.3032
-0.4000	0.3446	0.3704	0.3527	0.3783	0.3896
-0.2000	0.4207	0.4537	0.4376	0.4641	0.4761
0.0000	0.5000	0.5356	0.5220	0.5468	0.5582
0.2000	0.5793	0.6123	0.6019	0.6226	0.6338
0.4000	0.6554	0.6812	0.6747	0.6892	0.7006
0.6000	0.7257	0.7411	0.7390	0.7456	0.7585
0.8000	0.7881	0.7918	0.7944	0.7924	0.8074
1.0000	0.8413	0.8340	0.8408	0.8309	0.8479
1.2000	0.8849	0.8688	0.8784	0.8629	0.8810
1.4000	0.9192	0.8975	0.9083	0.8898	0.9073
1.6000	0.9452	0.9211	0.9319	0.9127	0.9285
1.8000	0.9641	0.9405	0.9500	0.9323	0.9452
2.0000	0.9772	0.9562	0.9636	0.9490	0.9581
2.2000	0.9861	0.9687	0.9737	0.9627	0.9680
2.4000	0.9918	0.9784	0.9812	0.9738	0.9756

distribution. For illustrative purposes, we show some results for the chi-squared distribution with 4 degree of freedom. This is a particular Gamma distribution with the density $f(x) = \frac{1}{4}xe^{-x/2}$, $x > 0$. After changing variables, we see that for this distribution, $Q(p)$ is a solution of the equation:

$$p = \int_0^{Q(p)/2} ye^{-y} dy, \quad (3.20)$$

which means that $Q(p)$ can be expressed via the inverse of an incomplete gamma function. The derivatives of $Q(p)$ are then easily obtained by using the $Q(p)$ value. Indeed, the following calculations follow easily by applying integration by parts in (3.20) and differentiating both sides of the equality with respect to p :

Table 3.3 Chi-square sample

v	Normal	Edgeworth	True	Edgeworth	True	Edgeworth	True
−2.4000	0.0082	0.0007	0.0026	−0.0005	0.0009	−0.0042	0.0004
−2.2000	0.0139	0.0042	0.0059	0.0026	0.0027	−0.0002	0.0016
−2.0000	0.0228	0.0111	0.0124	0.0092	0.0069	0.0036	0.0053
−1.8000	0.0359	0.0231	0.0238	0.0209	0.0154	0.0147	0.0138
−1.6000	0.0548	0.0419	0.0423	0.0397	0.0306	0.0334	0.0304
−1.4000	0.0808	0.0695	0.0698	0.0675	0.0553	0.0620	0.0589
−1.2000	0.1151	0.1074	0.1079	0.1059	0.0915	0.1021	0.1008
−1.0000	0.1587	0.1563	0.1575	0.1557	0.1406	0.1544	0.1576
−0.8000	0.2119	0.2162	0.2185	0.2166	0.2022	0.2185	0.2271
−0.6000	0.2743	0.2857	0.2888	0.2871	0.2745	0.2924	0.3061
−0.4000	0.3446	0.3622	0.3658	0.3646	0.3545	0.3729	0.3903
−0.2000	0.4207	0.4427	0.4470	0.4457	0.4385	0.4561	0.4750
0.0000	0.5000	0.5235	0.5281	0.5268	0.5226	0.5379	0.5571
0.2000	0.5793	0.6012	0.6061	0.6043	0.6025	0.6147	0.6332
0.4000	0.6554	0.6731	0.6782	0.6755	0.6762	0.6838	0.7010
0.6000	0.7257	0.7372	0.7427	0.7386	0.7417	0.7439	0.7602
0.8000	0.7881	0.7925	0.7986	0.7929	0.7981	0.7948	0.8101
1.0000	0.8413	0.8390	0.8456	0.8384	0.8447	0.8371	0.8513
1.2000	0.8849	0.8772	0.8836	0.8757	0.8827	0.8719	0.8848
1.4000	0.9192	0.9080	0.9139	0.9060	0.9129	0.9005	0.9118
1.6000	0.9452	0.9323	0.9375	0.9301	0.9364	0.9238	0.9329
1.8000	0.9641	0.9512	0.9555	0.9490	0.9541	0.9428	0.9493
2.0000	0.9772	0.9656	0.9687	0.9637	0.9673	0.9581	0.9619
2.2000	0.9861	0.9764	0.9782	0.9748	0.9770	0.9702	0.9716
2.4000	0.9918	0.9843	0.9852	0.9831	0.9840	0.9794	0.9790

$$Q'(p) = \frac{4 + 2Q(p)}{(1 - p)Q(p)}, \quad Q''(p) = \frac{2Q(p)(2 + Q(p)) - 4Q'(p)(1 - p)}{(1 - p)^2 Q(p)^2}$$

In this case, the improvement due to the Edgeworth approximation is significant and can be felt for sample sizes as small as $n = 15$. Again, the improvement seems to be virtually uniform all over the whole range of values of the argument (except for a small region in the very left tail where the Edgeworth expansion may become negative, in which case it should be equal to zero). The numerical values of “true” cumulative distribution function of the standardized random variable $\sqrt{n}(\hat{Q}_{p,h} - Q(p))/\sigma_n$ are compared to the cumulative distribution of the standard normal and to the Edgeworth in Table 3.3. The same number of simulations has been applied as in the previous examples. In order to demonstrate the increasing accuracy of both the Edgeworth and the normal approximation with increasing sample size, as well as the uniform

improvement offered by the Edgeworth expansion for large n such as $n = 50$, the numerical values are given for three different scenarios: $n = 50$, $n = 30$, and $n = 15$, with $p = 0.9$. Again, it can be seen that the improvement is virtually over the whole range of values of the argument and that the gains, especially for $n = 15$, are impressive.

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